

Creating the Sliding Number Game in App Inventor 2



If you've been faithfully following this series, then get ready to create a fun game. This is the popular sliding number puzzle which contains numbered tiles within a square frame, with one blank space for a tile. The idea is to move a tile to the blank space, and rearrange the tiles in sequence in the shortest possible time and with the least number of moves.

We started Android app development with App Inventor, picking up some pretty basic ideas to get used to the concepts, and we have moved towards more challenging and engaging application ideas.

I have been getting regular feedback, suggestions and debugging questions. By now, most readers have gained enough experience and hands-on practice with all the features of the tool. If you are a first-time reader of this series, you too can very well learn from this point onwards without having any prior programming experience or any specialised knowledge. Apps developed with App Inventor can be uploaded to the Google Play Store and I will be delighted to see the good work done by readers.

If we talk about App Inventor specifically, then we can proudly say that we have mastered the components

that are available in the palette, and moving between designer and block editor is so much fun.

In this tutorial, I will give you the rather challenging task of building yet another interesting and powerful game with the resources available in App Inventor 2. I will be making a 15-number sliding puzzle which is also known by other names such as Gem Puzzle, Boss Puzzle, Game of Fifteen, Mystic Square, etc. It also comes in various sizes and variations. The objective is to arrange the numbers in sequence, leaving only the corner block empty to allow the number blocks to be moved around. If the size of the puzzle is 3×3 tiles, it is called the 8-puzzle or 9-puzzle, and if it is made up of 4×4 tiles, it is called the 15-puzzle or 16-puzzle—named, respectively, for the number of tiles or the number of spaces.

Theme of the application

The theme is pretty simple and you

probably have already got the idea from descriptions in earlier articles in this series. We will make an Android application that will present a square board, which is covered with numbers on tiles (in a series starting from 1 but in jumbled order) with one empty space. The objective is to arrange the numbers in sequence. To make it more interesting, you can count the time and the number of moves it takes you to solve the puzzle so that you can challenge your friends as well.

GUI requirement

For every application, we have a graphical user interface or GUI, which helps the user to interact with the on-screen components. The block editor section defines how each component responds to user actions.

GUI requirements for Screen1

1. **Label:** This is a static text component that is used to display